An Application Hub for Offline Clinical Data Collection in Neurological Field Brigades

Offline integration for clinical field brigades · **IEEE COLCOM 2026**

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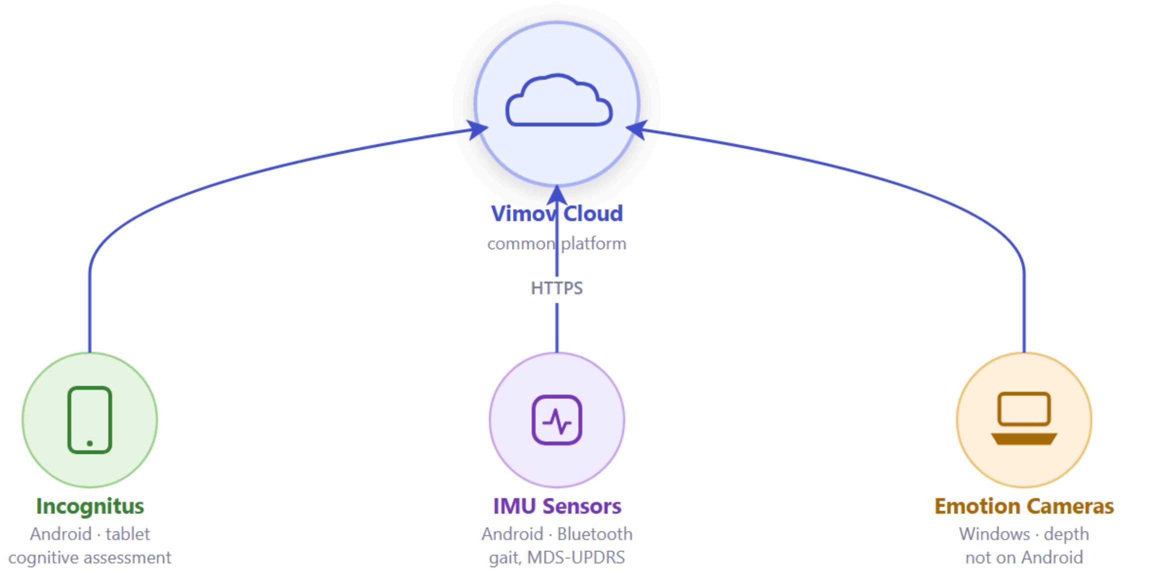
The context · LargePD field brigades

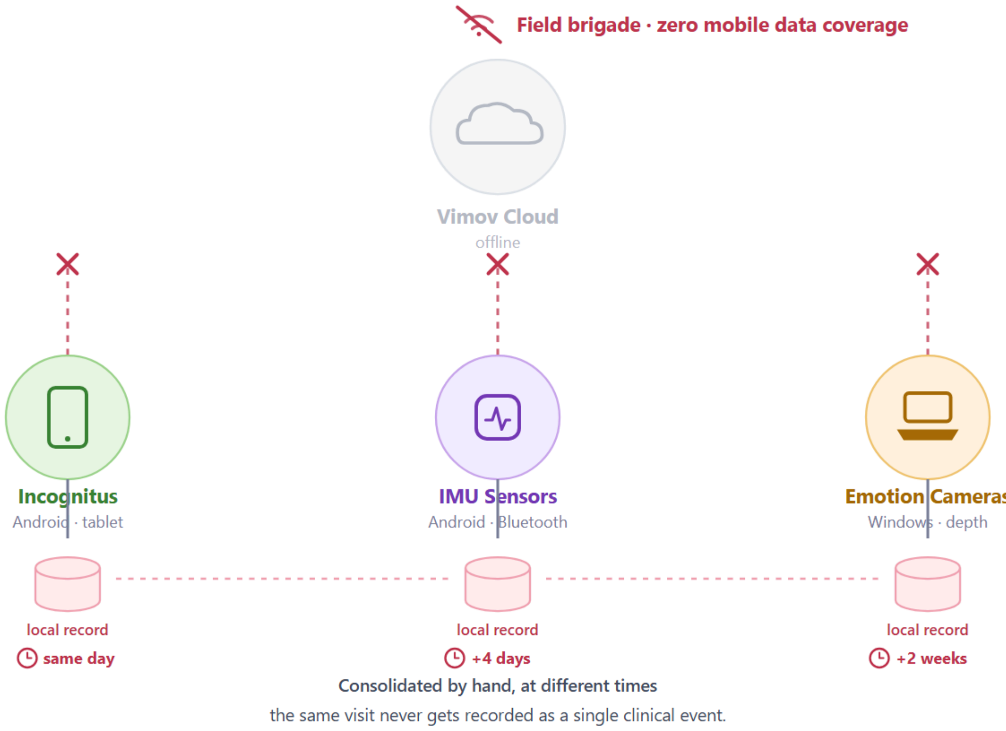
- The i2t group (Universidad Icesi) and the Neurology department at Fundación Valle del Lili are running **LargePD**, a Parkinson's disease study in southwestern Colombia
- The brigades already cover 7 sites (Tumaco, Jamundí, Santander de Quilichao, among others) and assess close to 150 patients on each visit
- At those sites connectivity is usually limited or nonexistent · exactly the scenario the three apps don't account for, since all of them assume internet is always available

Three applications, three different worlds



Three different teams, three different stacks · all talk to the same platform when there's internet.

The real problem



How can offline operation be achieved without touching any of the three apps?

State of the art · the gap

- Lomotey et al. (2016, 2017 MUHIS, 2018) and Agarwal (fault tolerance) assume the client app can be built or rebuilt around a sync SDK
- That assumption doesn't hold here: all three apps are already in production and cannot be modified
- Conclusion: no prior work covers local multi-app integration together with "zero client modification"

Comparison with related work (Table I)

No prior work covers local multi-app integration without modifying the client

Requirement	[2]	[3]	[4]	[5]	This work
Local multi-app integration	–	–	–	–	✓
Local persistence	✓	✓	✓	✓	✓
Automatic remote sync	✓	✓	✓	✓	✓
Heterogeneous devices	–	~	–	~	✓
Mobile brigade portability	✓	–	~	–	✓
Zero client modification	–	–	–	–	✓

✓ Yes ~ Partial – No

Only row where every other approach is "–"

Qualitative comparison against related work

Scope of the contribution

The individual building blocks (edge computing, store-and-forward) are well known · the contribution is one of **systems integration**, deliberately scoped. The hard constraint — zero client-side changes — is what forces the integration to move from the application layer down to the network layer.

3-layer interception

without touching clients

Store-first middleware

unified patient identity

Quantitative comparison

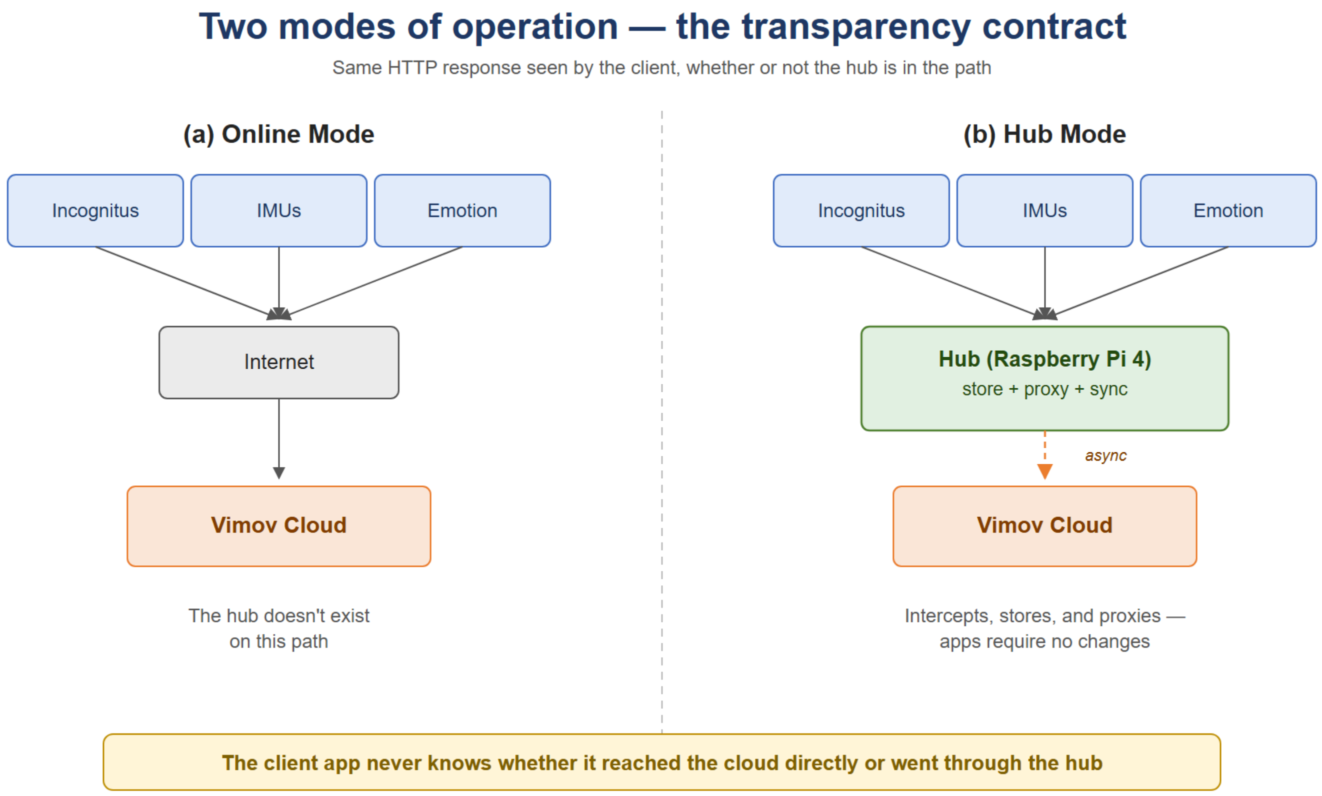
against real baselines

Lab + field-brigade validation

explicit security posture

Architecture · overview, two modes

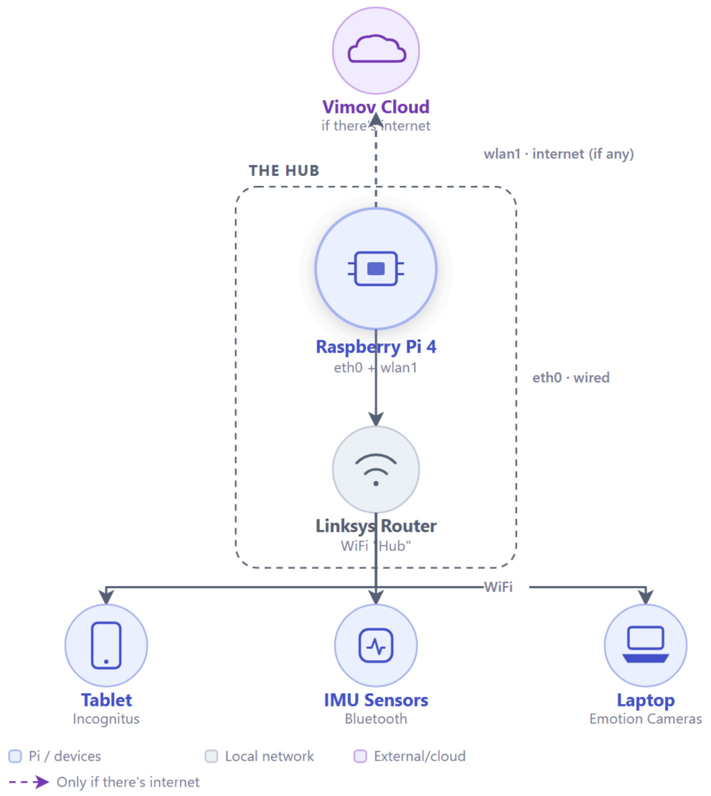
- **Online mode:** the app talks directly to Vimov Cloud · normal behavior
- **Hub mode:** local DNS resolves the domain to the Pi itself, which intercepts, stores, and proxies
- The "transparency contract": the app sees exactly the same HTTP behavior whether or not the hub is in the path



Two modes of operation: direct to the cloud, or intercepted by the hub

Physical topology in the field

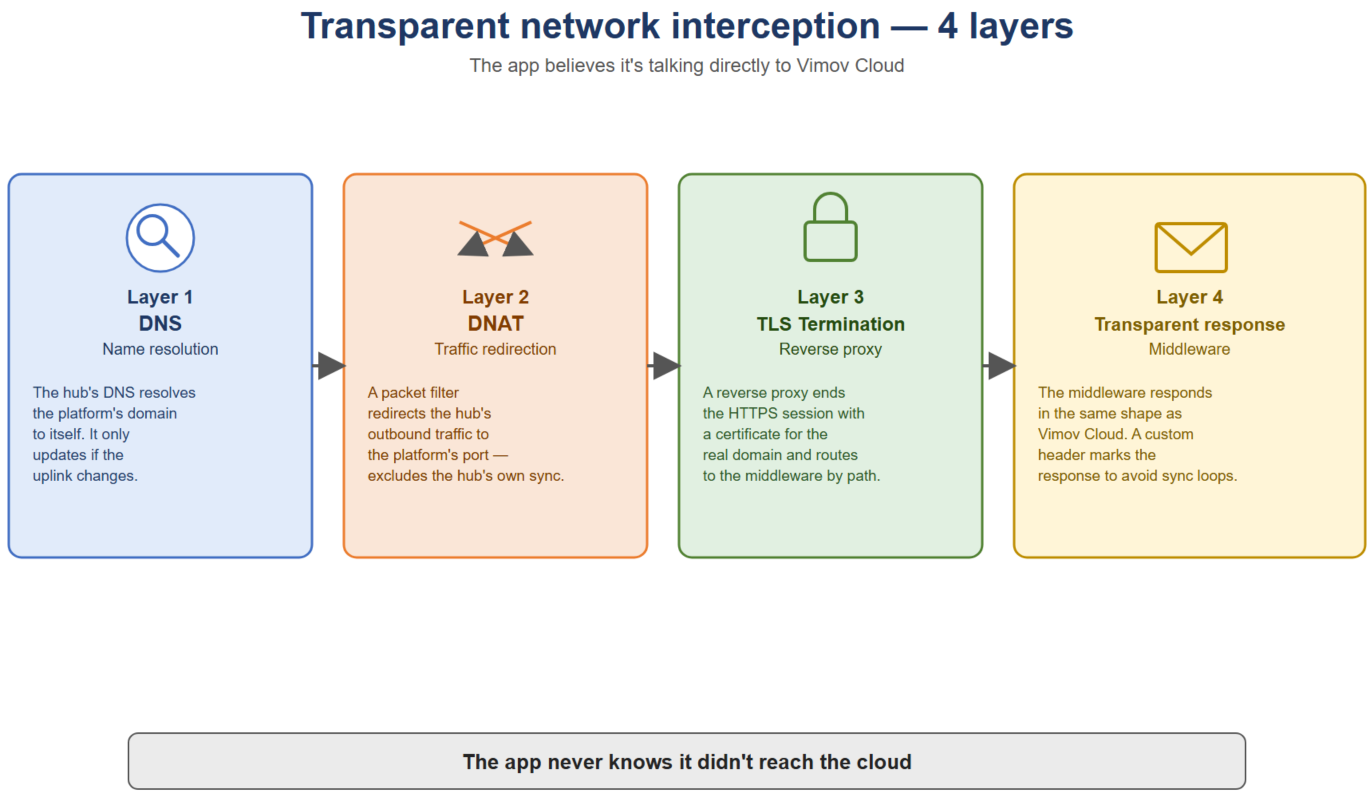
- Raspberry Pi 4 (8 GB) with two interfaces: LAN to the brigade router, WiFi uplink to internet/campus
- Three different client devices, connected to the router's WiFi · not directly to the Pi
- Three possible boot modes (internet / ap-extension / standalone) · **ap-extension** is the actual field-brigade mode



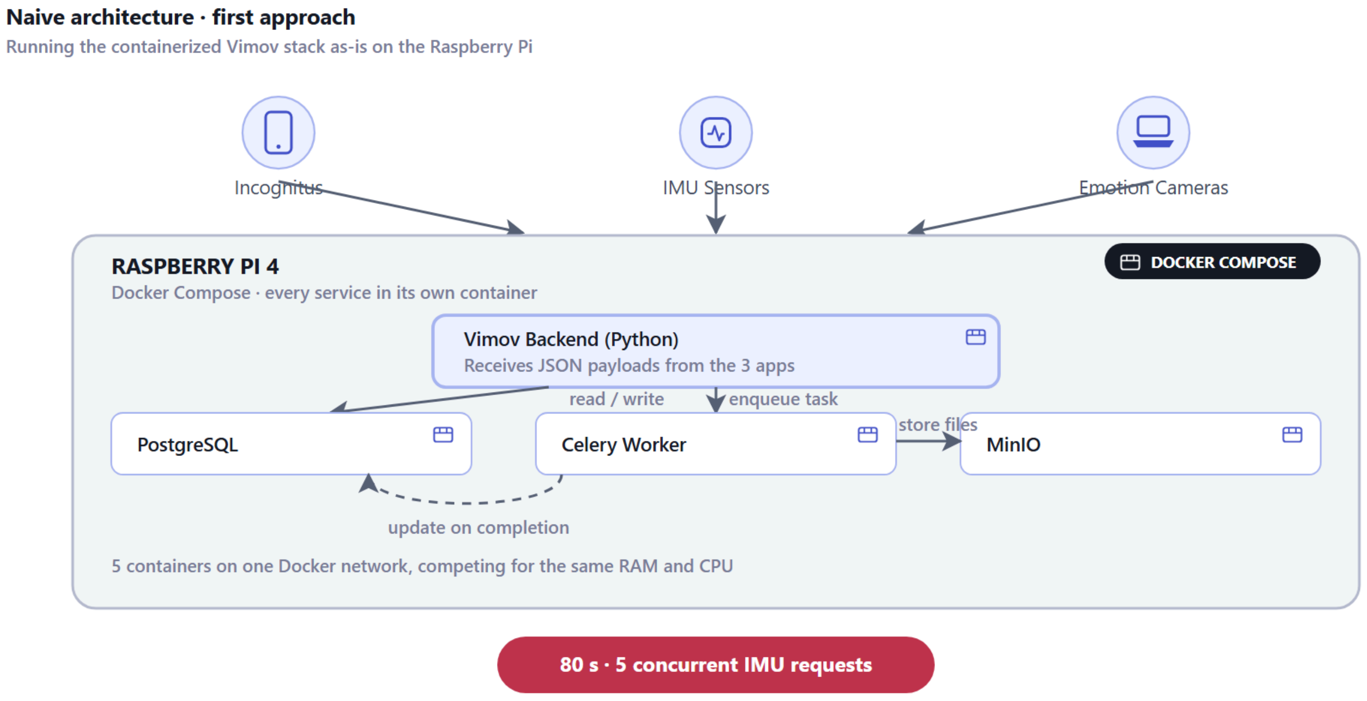
Transparent network interception · 4 layers

- **Local DNS** · resolves the real domain to the Pi itself
- **DNAT** · redirects incoming traffic to the right port
- **TLS** · nginx terminates with the real domain's certificate, embedded in the Pi image
- **Transparent response** · structurally identical to Vimov Cloud, with an anti-loop header

Key takeaway: the app never knows it didn't reach the cloud.



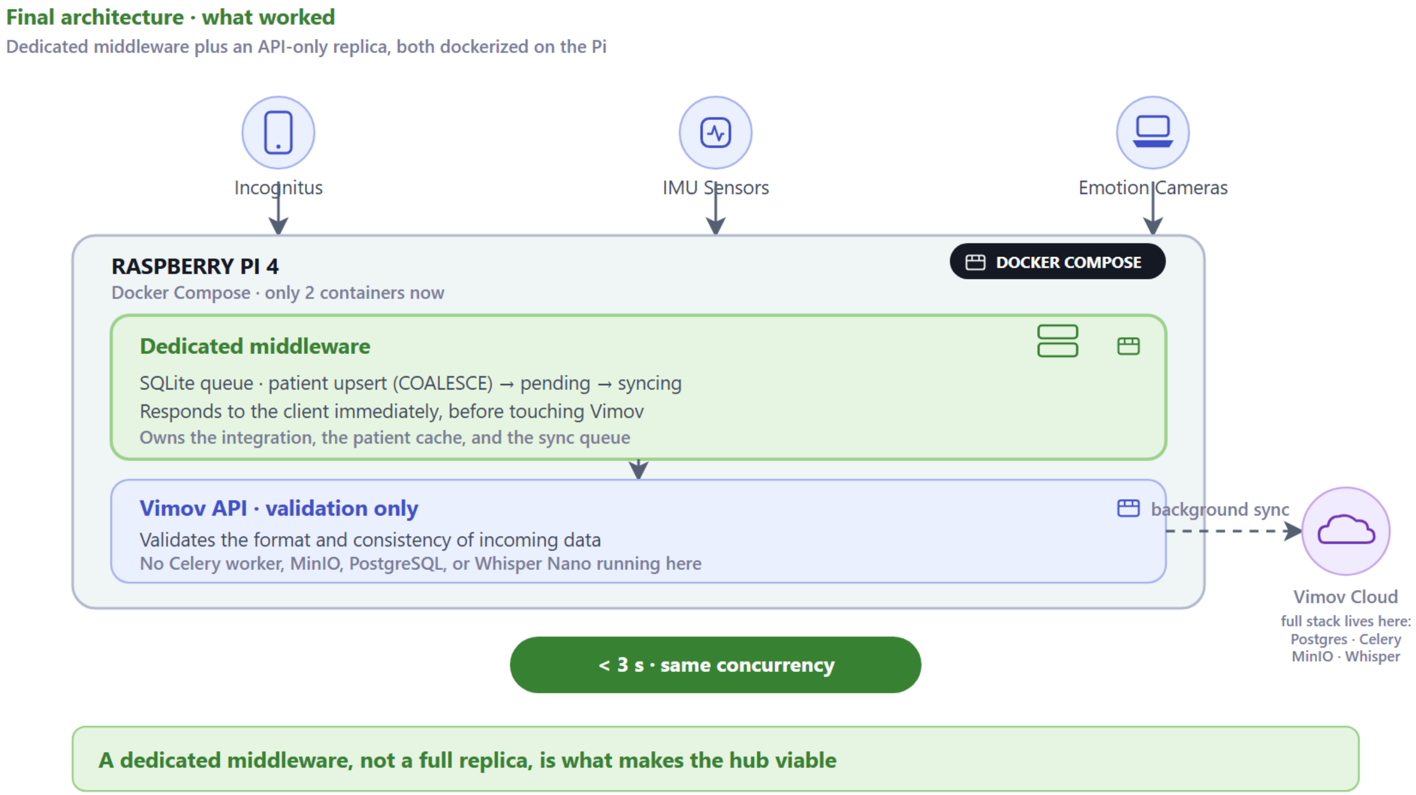
Naive architecture · first attempt



What we thought first: concurrency was blamed

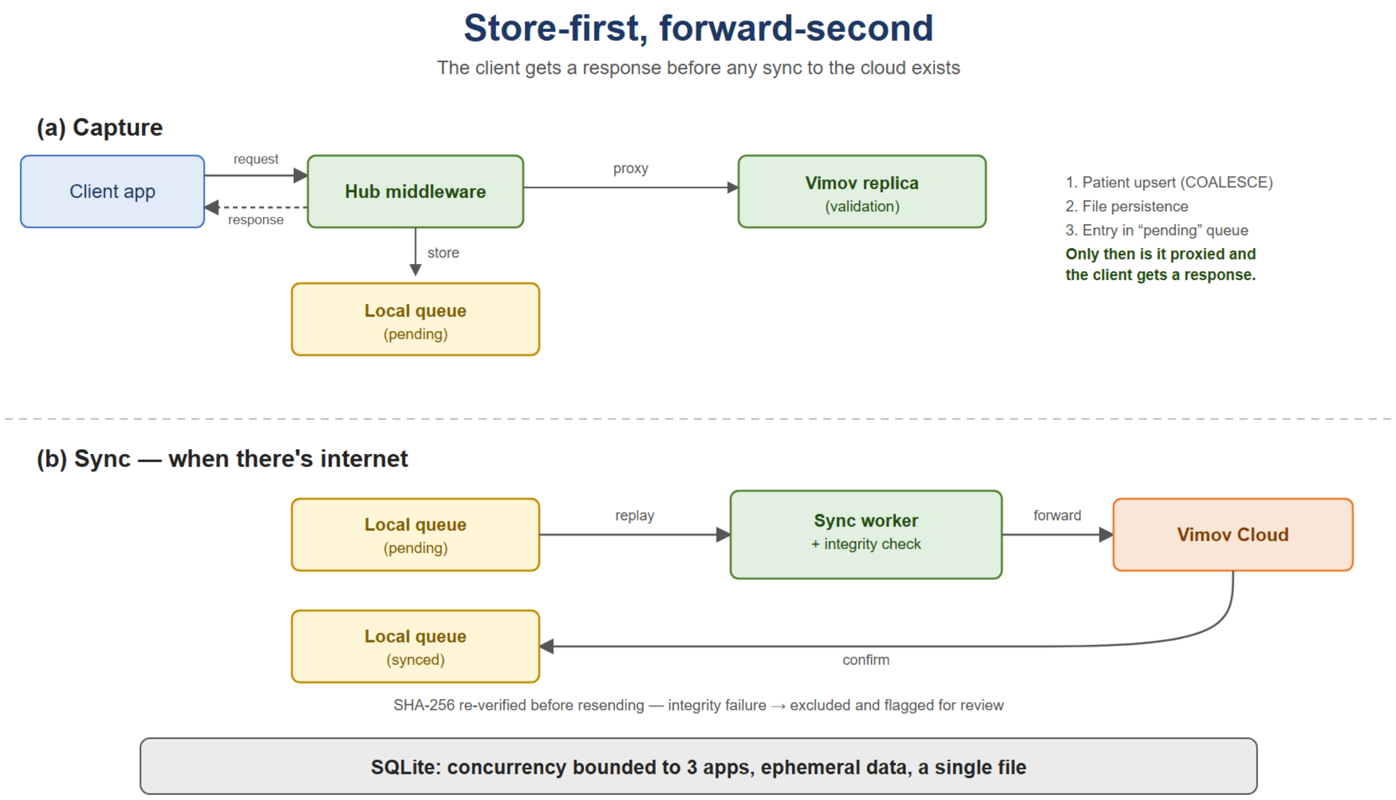
Real cause: duplicate processing over shared resources on the same Pi ·
Whisper Nano (speech-to-text for one cognitive test) also weighed on RAM, not just the request volume

Final architecture · what worked



Pipeline store–first, forward–second

- Three atomic steps before touching Vimov: **patient upsert** (COALESCE, never overwrites with null), file persistence, "pending" queue entry
- Then it's proxied and the client gets a response · syncing to the cloud happens afterward, in the background
- SQLite for the middleware, not Postgres: concurrency is bounded to 3 apps, data is ephemeral, a single file to back up or clear



Local capture followed by sync to the cloud

DNS + TLS · why the interception is transparent

Local DNS

- The 3 apps are hardcoded to **Vimov Cloud's real domain** · one hostname, three independently developed clients
- The Pi's local DNS resolves that domain **to itself** · every request from Incognitus, IMU Sensors, and Emotion Cameras is captured, with zero client-side changes

TLS termination

- Vimov Cloud is **HTTPS-only**, so the apps expect a valid TLS session for that exact domain
- nginx on the Pi terminates TLS with a certificate for the real domain, **embedded in the Pi image** · never published to a public trust store

Key takeaway: one DNS override + one embedded certificate route the full request set from all 3 measurement devices through the hub — transparently, with no client-side code changes.

Results, in three parts

Concurrency

80 s → <3 s

Response time when the 3 apps hit the hub at once · naive vs. final architecture, same hardware

Load

132 / 132

Requests with no errors, across sustained throughput, bulk sync, and a stress test on the Pi's RAM and CPU

Coverage

11.9 m → 20.6 m

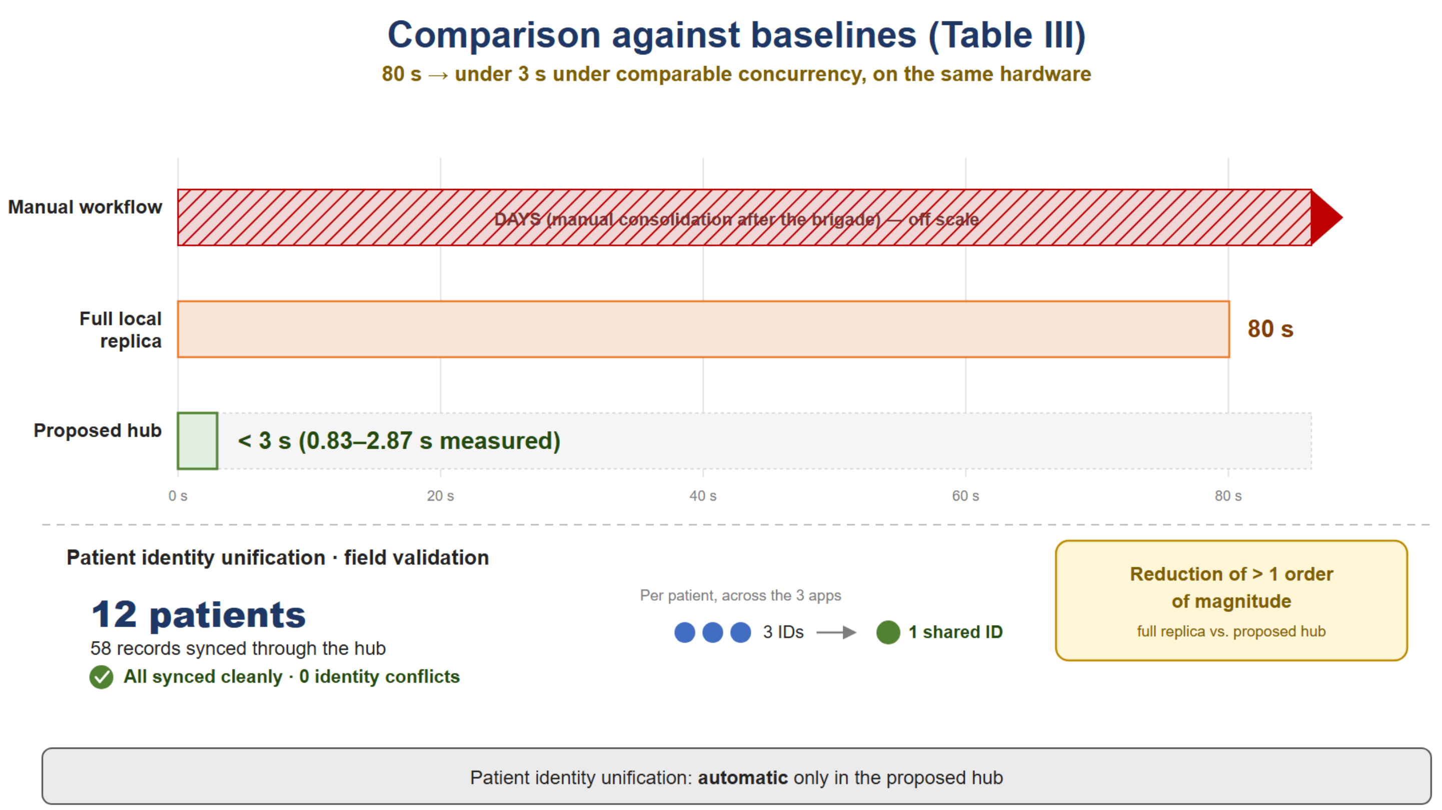
WiFi range in the field · the Pi's built-in antenna vs. an external access point

Followed by combined field validation, at two sites

CONCURRENCY

Results · comparison against baselines

- **80 s → under 3 s** under comparable concurrency, on the same hardware
- Patient identity entries: 3 (manual workflow) → 3 (local replica) → **1** (proposed hub)
- Confirms that the dedicated middleware, not the full replica, is what makes the hub viable

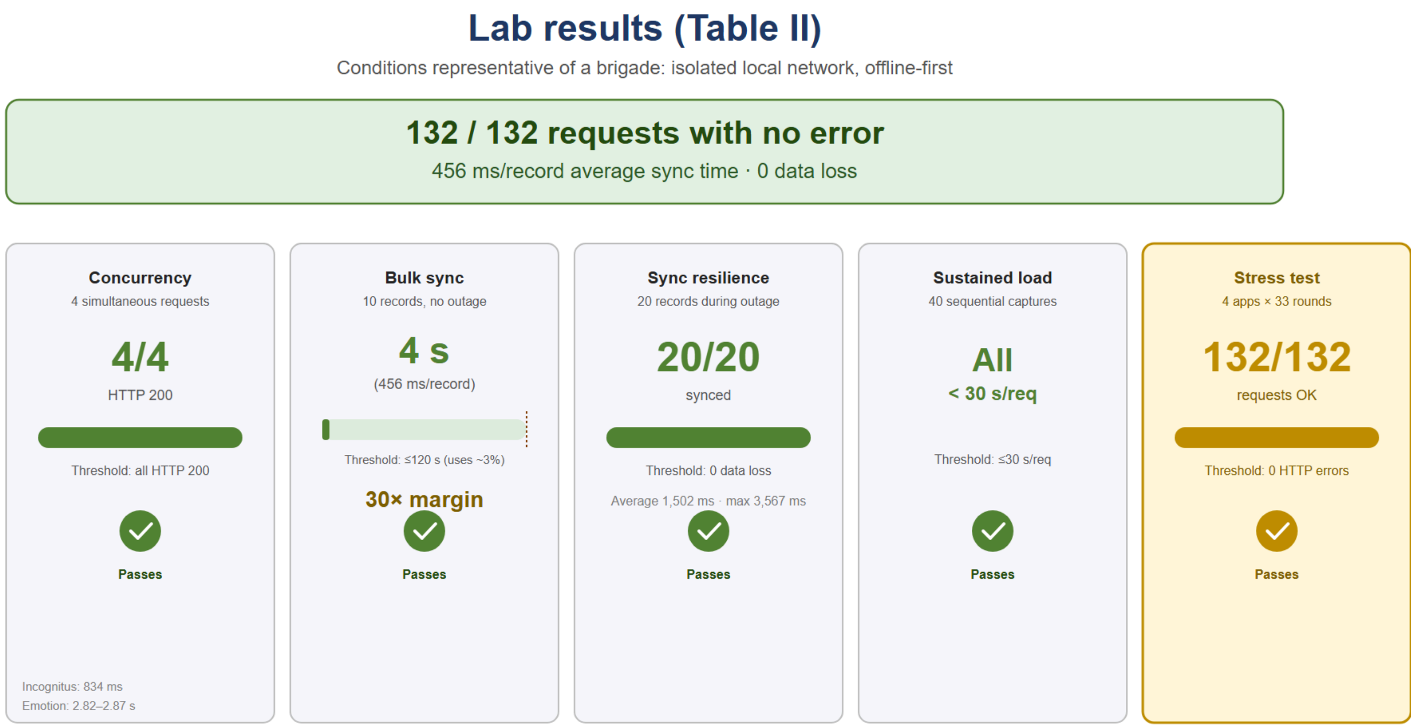


Time comparison against the alternatives

LOAD

Results · load on the Pi

- **132/132 requests** with no errors, across concurrency, bulk sync, and a 4-apps-×-33-rounds stress test
- **456 ms** per record, average sync time
- **20/20 records** recovered after a simulated power outage · 0 data loss under sustained load



Lab test results

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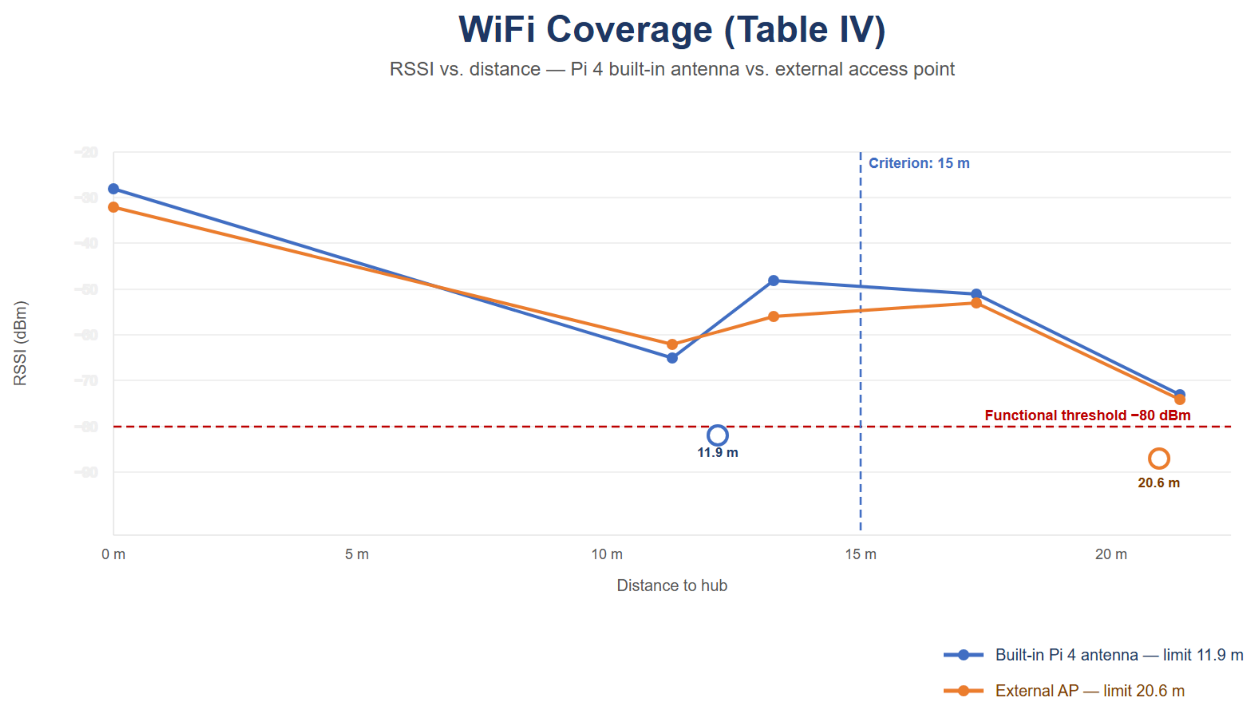
COVERAGE

Results · WiFi coverage

- Pi's built-in antenna: functional limit of **11.9 m** (below the 15 m target)
- With an external AP: **20.6 m** · this is what motivated *ap-extension* mode as the recommended setup



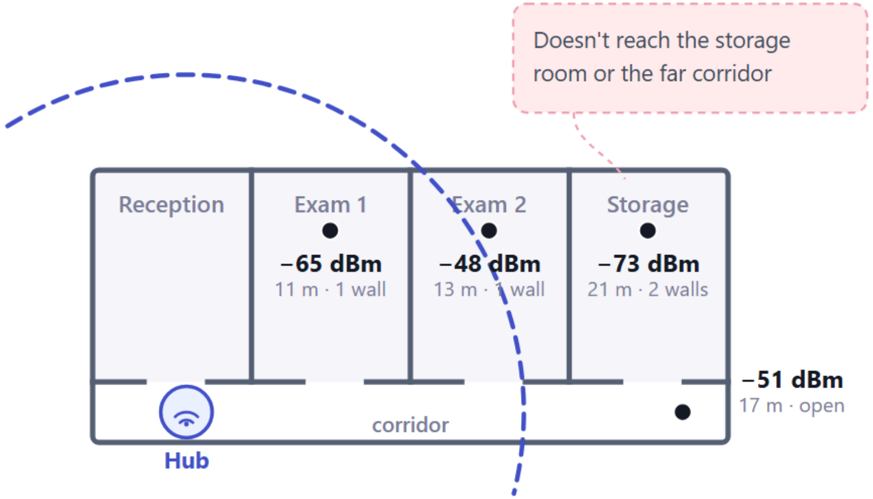
External AP motivated the ap-extension mode as the recommended configuration

COVERAGE

WiFi coverage · floor plan

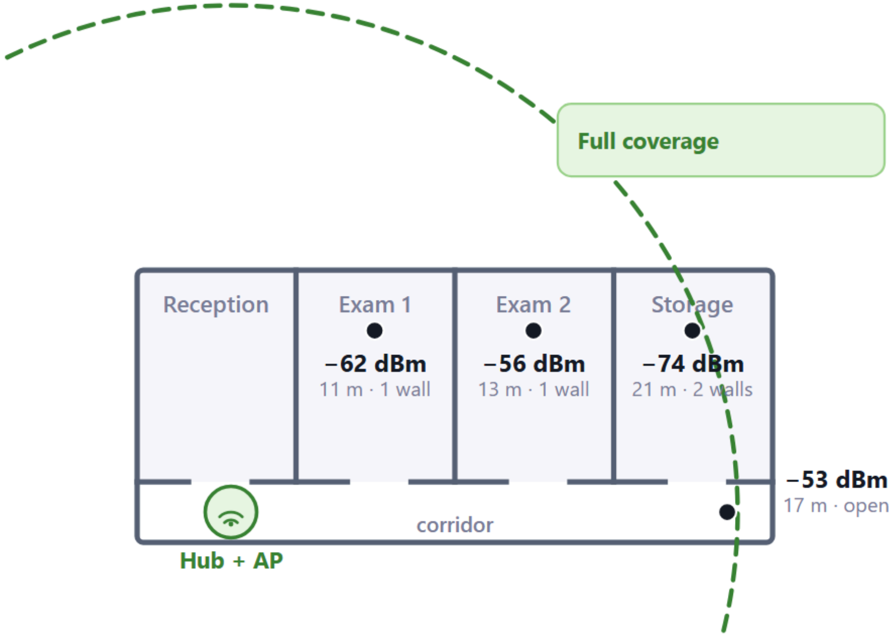
Built-in Pi antenna

Functional limit 11.9 m · below the 15 m criterion



+ External AP (ap-extension mode)

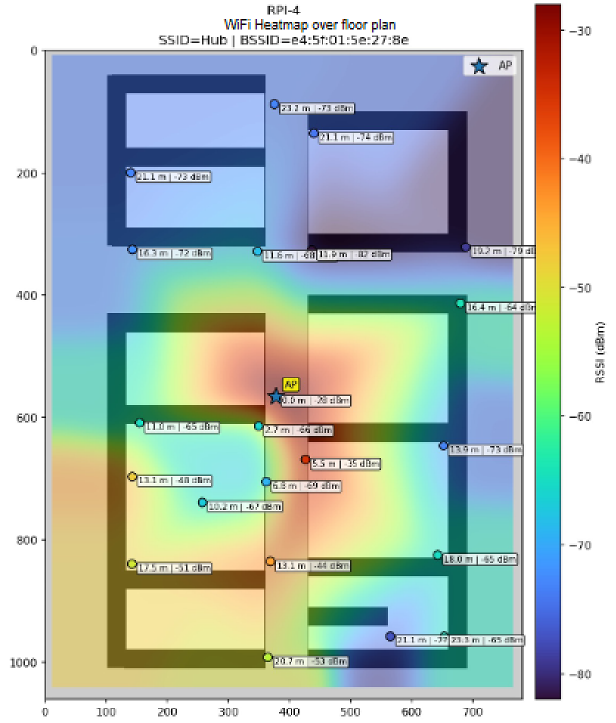
Functional limit 20.6 m · exceeds the 15 m criterion



Wall (solid line) · functional limit < −80 dBm (dashed line) · actual measurement point (RSSI)

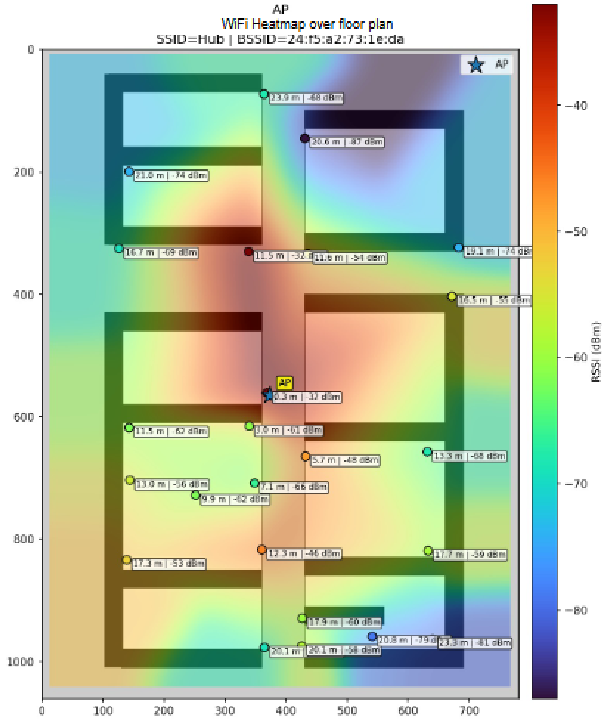
COVERAGE

WiFi coverage · real field heatmap



Pi built-in antenna (RPI-4)

Threshold ≥ −80 dBm · up to 11.9 m



+ external AP

Threshold ≥ −80 dBm · up to 20.6 m

FIELD VALIDATION

Validation · two sites

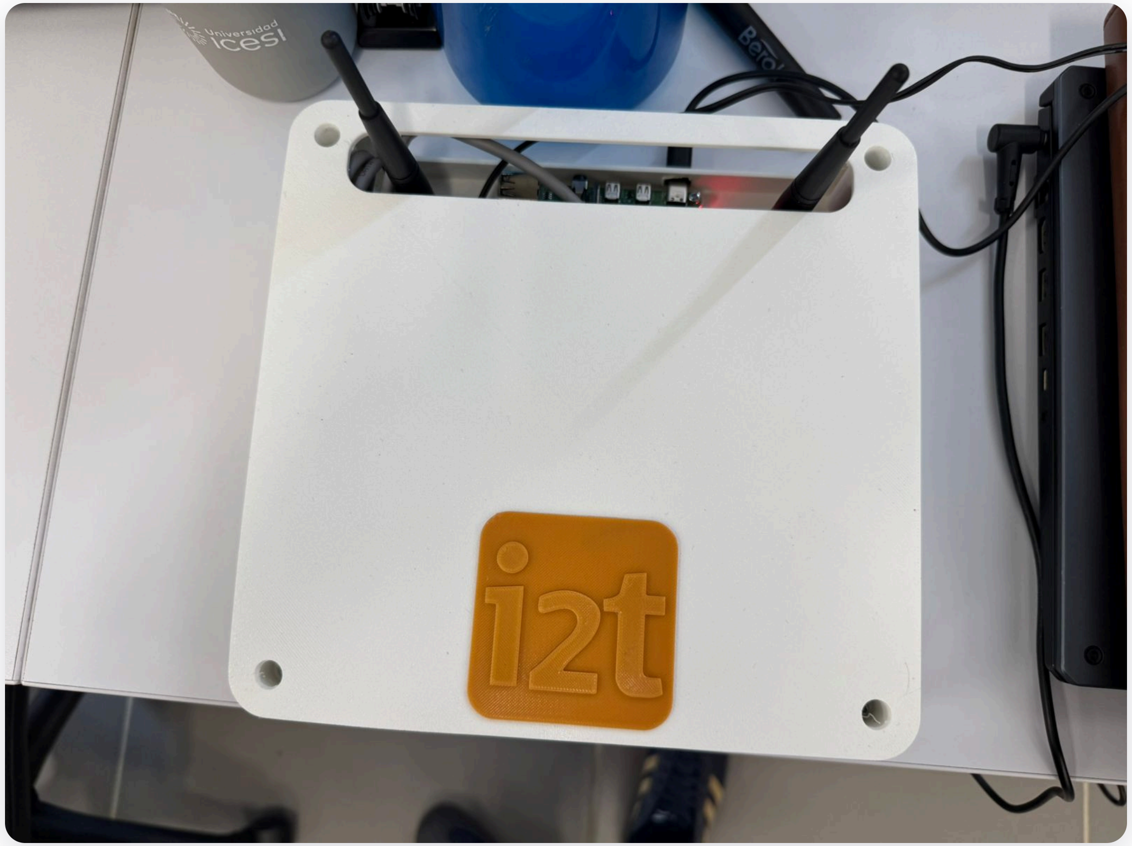
- **Research lab:** controlled load tests · 132/132 requests with no errors, full recovery after a simulated power outage
- **Fundación Valle del Lili**, real field brigade: 12 patients, 58 records that went through the hub · all **3 of 3 apps** tested in the field (Incognitus, IMU sensors, Emotion Cameras)
- Only field finding: NAT on wlan1 required a tweak for Emotion Cameras

Combined validation · two sites

2	12	58	3/3
sites	patients	records	apps in the field

132 additional requests validated under controlled lab load

The hub device



The hardware went through an enclosure design prototype to make it portable enough for a field brigade

Limitations and future work

- Field validation of **feasibility**, not scale (12 patients, 58 records vs. ~150 patients/brigade typical) · the lab does cover the expected order of magnitude in throughput
- Last-write-wins is reasonable for demographic data, not for clinical measurements · the current design doesn't distinguish between the two
- Future work: instrumented apps (pending PR), encryption at rest, per-user authentication on the panel, zero-touch provisioning, multitenancy

Conclusion

This isn't a new algorithm: it's proof that a fleet of independent, already-deployed clinical apps can become offline-capable without touching a single line of client code, unifying patient identity, on under $100 of hardware.

- A reusable edge-computing pattern for multi-app clinical integration in disconnected environments

Thank you!

Questions?

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